# lit-acquisition: An Open Multi-source Multilingual Biomedical Literature Acquisition System with an Exploratory Coverage Benchmark

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## ABSTRACT

Variant classification in clinical genetics (e.g., ACMG/AMP guidelines) requires clinicians to assemble evidence from biomedical literature that is scattered across many databases, publishers, preprint servers, and languages. We present **lit-acquisition**, an open-source pipeline that integrates 18+ heterogeneous literature providers — PubMed, EuropePMC, Crossref, OpenAlex, Semantic Scholar, preprint servers, language-specific databases (CiNii, J-STAGE, SciELO), and others — behind a single acquisition interface, with LLM-based cross-lingual query translation, citation graph traversal, and an open-access-aware download layer.

We report an exploratory coverage evaluation on a heuristically constructed reference set of 30 genetic variants (540 papers) that was *not* verified by domain experts and whose candidate pool overlaps the evaluated system; all results are therefore relative coverage comparisons, not clinical retrieval quality claims. Under a strictly nested ablation protocol (each configuration extends a frozen candidate list of the previous one) and Holm-corrected paired tests, the end-to-end pipeline roughly doubles PubMed-only set recall (0.171 → 0.345; Holm-adjusted $p$=0.004, rank-biserial $r$=0.79). Individual component deltas are positive but not significant after correction on 30 variants (multi-source +0.076, translation +0.082, citation graph +0.016). In a controlled same-provider contrast, query translation recovers 21 of 24 annotated German and French papers that English-only configurations miss entirely; claims for other languages remain unevaluated. Citation traversal contributes 11 relevant papers — 10 of which appear in no keyword-channel list — concentrated in 3 of 30 variants, at a significant list-level precision cost (−0.058). We further quantify live-API run-to-run variance, report transparent single-source and fusion baselines, and release the toolkit, reference set, raw candidate lists, and a deterministic analysis pipeline for reproducible re-evaluation. The relevance-gating and download layers are described as engineering features and are not experimentally validated here.

## CCS CONCEPTS

• **Information systems** → **Information retrieval**; *Search engine indexing*; • **Computing methodologies** → *Natural language processing*.

## KEYWORDS

biomedical literature, multilingual retrieval, citation graph, cross-lingual IR, variant classification, ACMG/AMP

## 1 INTRODUCTION

Interpreting a genetic variant for clinical reporting requires assembling published evidence: case reports, functional studies, sequencing cohorts, and guidelines that establish whether a variant is pathogenic or benign. Under the ACMG/AMP guidelines [1], this evidence supports criteria such as PS3 (well-established functional effect) and PVS1 (null variant). Evidence is scattered across publisher databases (PubMed, Crossref, EuropePMC), preprint servers (arXiv [22], bioRxiv, medRxiv), regional and language-specific repositories (CiNii [24], J-STAGE [25], SciELO [26]), and trial and data registries (ClinicalTrials.gov [31], Zenodo [30]). Clinicians and genetic counselors routinely search several of these systems manually, deduplicate results by hand, and risk missing relevant non-English literature.

Commercial tools such as Mastermind [5] and systematic-review platforms (Covidence [6], Rayyan [7]) address parts of this workflow, but they are closed-source, cover limited sources, and do not provide automated open-access full-text retrieval with explicit license awareness.

We present **lit-acquisition**, an open-source, MIT-licensed toolkit for multilingual biomedical literature acquisition. This paper makes the following contributions, and we deliberately separate *implemented system capabilities* from *experimentally evaluated claims*:

(1) **System (engineering).** A unified acquisition pipeline integrating 18+ heterogeneous data sources with health-aware parallel search, schema normalization, multi-key deduplication (DOI → URL → normalized title), LLM-based cross-lingual query translation with language-aware provider routing, citation graph traversal, and an OA-aware download layer. The relevance gate, download layer, and health-aware scheduling are described in Section 3 but are *not* experimentally validated in this paper.
(2) **Evaluation protocol and dataset.** An exploratory, heuristically constructed reference set (30 variants, 540 papers; Section 4.1) released together with raw candidate lists, a strictly nested ablation protocol that removes live-API re-retrieval confounds (Section 4.2), fixed-denominator metrics, Holm-corrected paired statistics, and a deterministic script that regenerates every number in this paper from the raw lists.
(3) **Empirical findings (relative to this reference set).** The end-to-end pipeline roughly doubles PubMed-only set recall with a Holm-significant difference; individual component deltas are positive but not individually significant on 30

variants; translation recovers 21/24 of the annotated German/French papers in a controlled contrast; citation traversal adds a small number of otherwise-unretrieved papers at a significant list-level precision cost. We also quantify run-to-run API variance and report single-source and fusion baselines.

## 2 RELATED WORK

**Biomedical literature search.** PubMed [2] and EuropePMC [3] are the de facto standard sources for biomedical literature; both expose search APIs with fielded queries. Semantic Scholar [4] provides a large-scale citation graph and paper-level embeddings. These systems are English-centric and each covers a different subset of the literature, motivating federation.

**Commercial evidence-acquisition tools.** Mastermind [5] is a proprietary genomic variant search engine that aggregates literature per variant; Covidence [6] and Rayyan [7] support systematic-review screening but are closed-source and not aimed at full-text acquisition with license handling.

**Federated / multi-source search.** Meta-search architectures such as MetaCrawler [8] established result aggregation from heterogeneous engines; in the scholarly domain, aggregators and directories such as OpenAlex [9], DOAJ [23], BASE [27], CORE [28], and OpenAIRE [29] expose broad metadata. Rank fusion methods such as Reciprocal Rank Fusion [20] combine per-source rankings without score calibration; we use RRF and simple union as transparent federation baselines. Our system treats these sources as one layer of an acquisition pipeline rather than a search UI, adding normalization, dedup, and download.

**Citation analysis in IR.** Bibliographic coupling [10] and co-citation [11] are classical similarity measures; PageRank [12] exploits citation topology for ranking. Literature-based discovery [13] showed that citation/reference edges carry unexploited relevance. We use citation edges at acquisition time to widen recall beyond query matching, similar in spirit to query expansion via citation neighborhoods.

**Cross-lingual IR.** Classic CLIR translates queries with bilingual dictionaries or MT [14]; modern LLMs [15, 16] make translation of short domain-specific queries practical with minimal setup, which we exploit.

**Evaluation of IR systems.** Standard practice uses Recall@K, Precision@K, F1@K, and nDCG@K [17] with paired significance tests (Wilcoxon signed-rank [18]) and multiplicity control across test families (Holm [19]). We follow this protocol on a new, publicly released reference dataset (Section 4.1), and additionally report micro-averaged pooled counts and bootstrap confidence intervals of paired differences.

## 3 SYSTEM DESIGN

### 3.1 Architecture Overview

lit-acquisition is a three-phase pipeline (Figure 1): **(1) Acquisition** fans a query out in parallel across providers with health-aware prioritization; **(2) Normalization** unifies the heterogeneous schemas and deduplicates; **(3) Relevance Gate & Retrieval** ranks candidates, optionally traverses the citation graph, applies an LLM relevance gate, and downloads open-access full texts.

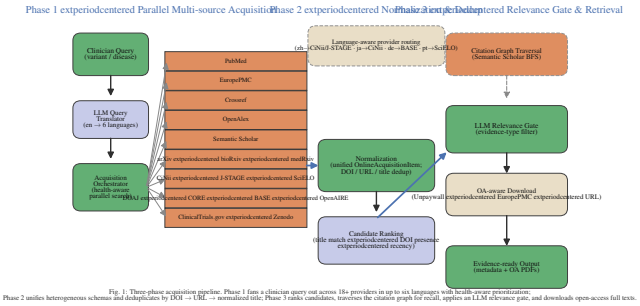



**Figure 1: System architecture: three-phase acquisition pipeline.**

**Table 1: Provider integrations: search/download capability and license behavior.**

| Provider | Search | Download | Notes |
|---|---|---|---|
| PubMed (E-utilities) | ✓ | — | English-centric |
| EuropePMC | ✓ | ✓ | OA + PMC via PMCID |
| Crossref | ✓ | — | DOI registration metadata |
| OpenAlex | ✓ | — | Open scholarly index |
| Semantic Scholar | ✓ | ✓ | Citation graph, TLDRs |
| arXiv | ✓ | ✓ | Preprints |
| bioRxiv / medRxiv | ✓ | ✓ | Preprints, CC-BY/CC0 |
| DOAJ | ✓ | — | OA journals |
| J-STAGE / CiNii | ✓ | — | Japanese literature |
| SciELO | ✓ | — | Latin American OA |
| BASE / CORE / OpenAIRE | ✓ | — | OA aggregators |
| ClinicalTrials.gov | ✓ | — | US public-domain trials |
| Zenodo | ✓ | ✓ | CERN OA, CC0 metadata |

✓ = supported; — = not provided.

### 3.2 Multi-source Search

The acquisition orchestrator (Section 3.1, implemented in `workflow.py`) issues parallel searches to the providers listed in Table 1. Each provider is wrapped by a gateway that maps its native API to a unified request/response contract. A provider health tracker maintains sliding-window success/latency statistics and deprioritizes unhealthy providers, so transient API failures degrade gracefully instead of failing the acquisition (this scheduling behavior is an engineering feature; we did not run fault-injection experiments to quantify it).

### 3.3 Normalization and Deduplication

Providers return heterogeneous item schemas. A normalization layer maps each result into a unified `OnlineAcquisitionItem` with canonical fields (title, authors, journal, year, DOI, PMID, language, license, links). Deduplication merges candidates by DOI, then URL, then normalized title, in that order of reliability (Table 2): an incoming item is dropped if its DOI, URL, or punctuation-stripped lowercase title has already been seen. Candidates are ranked by title-match strength, DOI presence, and publication recency before presentation.

**Table 2: Multi-key deduplication procedure.**

**Input:** candidate items from all providers and web search.
**Output:** deduplicated candidate set.

1. $S_{doi} \leftarrow \emptyset$; $S_{url} \leftarrow \emptyset$; $S_{title} \leftarrow \emptyset$
2. **for** item in api_items:
3. $doi \leftarrow$ normalize(item.doi); $url \leftarrow$ item.url; $title \leftarrow$ norm_title(item.title)
4. **if** $doi \in S_{doi}$ **or** $url \in S_{url}$ **or** $title \in S_{title}$:
5. **continue** (*drop duplicate*)
6. add $doi, url, title$ to $S_{doi}, S_{url}, S_{title}$; append item
7. **for** link in web_search_links:
8. apply same keys against $S_{doi}, S_{url}$

**Table 3: Language-aware provider routing (excerpt).**

| Query language | Providers |
|---|---|
| en | PubMed, EuropePMC, Semantic Scholar, Crossref, arXiv, bioRxiv, medRxiv, OpenAlex, OpenAIRE, BASE, CORE, ClinicalTrials.gov, Zenodo |
| zh | Crossref, Unpaywall, OpenAlex, DOAJ, PMC |
| ja | J-STAGE, CiNii, Crossref, Unpaywall, DOAJ, PMC |
| es / pt | SciELO, Crossref, Unpaywall |
| de | Crossref, EuropePMC, Unpaywall, OpenAlex, BASE, DOAJ |
| fr | Crossref, EuropePMC, Unpaywall, OpenAlex, DOAJ, PMC |
| ru | PMC, EuropePMC, Crossref, Unpaywall, OpenAlex |
| auto | Crossref, Semantic Scholar, Unpaywall, OpenAlex, EuropePMC, ClinicalTrials.gov, DOAJ, PMC, Zenodo |

## 3.4 Cross-lingual Query Translation

The `query_translator` module uses an LLM with a domain-specific prompt (released with the code; decoding temperature 0.2) that instructs the model to translate the query while preserving gene symbols, variant nomenclature (e.g., `p.G12D`, `c.35G>A`), and disease terminology verbatim. Queries are translated into six languages (en, zh, ja, de, fr, ru). The language-aware routing matrix (Table 3) then selects a provider plan per language, e.g., Japanese queries additionally target J-STAGE/CiNii, Spanish and Portuguese target SciELO, and German and French target EuropePMC (German additionally BASE). Note that the `auto` plan used by the federated configurations in our evaluation does not include the PubMed E-utilities channel itself (PMC serves as the NLM-backed channel); we return to the practical consequences of this in Section 4.9.

## 3.5 Citation Graph Traversal

Beyond keyword search, `citation_graph.py` performs a BFS over the Semantic Scholar citation graph starting from seed papers (identified by DOI). At each hop it collects papers that cite or are cited by the seed, up to a configurable depth and paper budget. Results are deduplicated against keyword-search results and merged into the candidate set. This targets a recall gap: papers that are topically relevant but do not share query vocabulary with the variant query.

## 3.6 Download and Relevance Gate (Not Evaluated Here)

Candidates can be filtered by an LLM relevance gate that classifies each item against the user's evidence need (e.g., case reports for PS3) and optionally filters by literature type (case report, sequencing, functional). Download routing is license-aware: DOI resolves through Unpaywall [21] for OA PDFs, PMCID through EuropePMC, and direct URLs with HTML→PDF redirect handling. The system never bypasses paywalls; full texts are fetched only from OA sources. **Scope note:** the relevance gate was disabled throughout the evaluation in this paper, and we report no experiments on gating accuracy, download success rates, or license-detection correctness; these layers are engineering capabilities whose validation is future work.

# 4 EVALUATION

## 4.1 Reference Dataset and Its Limits

We constructed an exploratory reference dataset of 30 genetic variants (V001–V030) spanning cancer genetics (BRCA1, TP53, KRAS, EGFR, BRAF, APC, MLH1, PALB2), cardiovascular (LDLR, SCN5A, MYBPC3, KCNQ1, FBN1), neurological (MECP2, HTT, SNCA, APP), metabolic (CFTR, PAH, GBA), hematological (HBB, F8, DMD, G6PD), and other domains, covering ACMG evidence criteria PS3 (14 variants) and PVS1 (6 variants) plus 10 additional clinically actionable variants. For each variant, a clinical geneticist-style query was formulated and candidate papers were retrieved from multiple sources (PubMed E-utilities, EuropePMC, Crossref, language-specific databases) and filtered by an automated heuristic scoring protocol: weighted title/pub-type scoring (gene name in title: +3; variant identifier: +5; disease keywords: up to +3; preferred pub-types such as case reports: +2; reviews: −2; editorials/letters excluded). After removing one duplicate annotation (the same DOI recorded twice for one variant), the dataset contains **540 unique papers** (mean 18.0 per variant, range 4–32), with languages including English (288), German (13), French (11), and 228 records with unknown language metadata. Annotations record DOI, PMID, title, language, and evidence type.

**Construct validity.** This dataset is a *reference set for relative system comparison*, not a clinically certified gold standard, for two reasons that the reader should weigh before interpreting any number in this paper. First, annotations were produced by the automated keyword protocol above and have **not** been verified by domain experts; they contain an unknown rate of false positives (papers scoring high on keywords without actually bearing on the variant) and false negatives (relevant papers the pooling protocol never surfaced), and keyword scoring cannot distinguish papers that are merely *about* a gene–disease pair from papers usable as variant-level ACMG evidence (e.g., treatment trials or reviews versus functional or case-level evidence). Second, the candidate pool from which annotations were drawn was generated by retrieval channels that overlap the evaluated system's providers, creating a circular dependency: methods sharing channels with the pool generator are structurally favored, and recall against this reference set measures

agreement with a heuristic pooling protocol rather than coverage of the true relevant literature. Consequently all recall/precision figures here are *relative coverage* measurements under identical annotation conditions. Expert re-annotation with inter-annotator agreement is prerequisite to any clinical-quality claim and is future work (Section 5). Only German and French carry non-English annotations, so multilingual claims in this paper are restricted to those two languages.

## 4.2 Methods and Collection Protocol

We evaluate an ablation chain over the same 30 variants:

- **M1: PubMed-only** — baseline PubMed esearch, English query.
- **M2: Multi-source** — federated providers (`auto` plan), original English query, no LLM translation.
- **M3: +Translation** — M2 + LLM query translation (6 languages) with language-aware routing.
- **M4: +Citation graph (nested)** — the frozen M3 candidate list of each variant, extended by citation-graph candidates (depth 1, budget 20, both directions, seeded by the first DOI of the list), deduplicated against the list and appended after it.

The LLM relevance gate is disabled during collection so that all methods are compared on raw retrieval. M1–M3 were collected as independent live runs on 2026-08-12 (12:09–13:47 local time, sequential, single run per method) with a per-variant candidate budget of 20; the run log records 78 HTTP 429 rate-limit rejections (all from anonymous Semantic Scholar access) and 120 J-STAGE client parse failures that the health-aware scheduler absorbed. M1 additionally hit a client-side cap of 15 PubMed candidates per query (a limitation of our collection client, since removed), so realized budgets were 14.2 (M1), 20.0 (M2), and 19.3 (M3) candidates per variant; metrics that depend on candidate depth are interpreted accordingly (Section 4.3).

**Why a nested M4.** Our original collection also re-executed the keyword phase for the full pipeline before appending citation candidates. Because providers are live APIs, that re-run returned keyword lists that differed from the frozen M3 lists on 12 of 30 variants (rate limiting, provider availability, and ordering changes), so the M3-to-M4 comparison confounded the citation-graph component with re-retrieval variance — visibly so, since recall *decreased* on some variants, which is impossible for a genuinely nested extension evaluated at fixed $K$. We therefore construct M4 deterministically from the recorded raw lists: frozen M3 list + recorded citation-graph candidates (202 items, none duplicating M3 entries), which restores the protocol guarantee Recall@$K$(M4) $\geq$ Recall@$K$(M3) for every variant and every $K$ (verified programmatically). The recorded traversal seeds coincide with the first DOI of the frozen M3 lists for all 30 variants, so the appended candidates are exactly those the nested protocol would have requested. We retain the original independent re-run (denoted M4′) solely to quantify run-to-run variance (Section 4.6), and the released benchmark runner now implements the nested protocol natively with a uniform default budget of 50 candidates for all methods.

## 4.3 Metrics

For each method and variant we compute, over the ranked candidate list: Recall@$K$; **Precision@$K$ with the standard fixed denominator** $K$ (methods returning fewer than $K$ items are penalized for missing ranks); F1@$K$; and nDCG@$K$ [17], for $K \in \{5, 10, 15, 20, 50\}$. Because realized candidate budgets were below 50 for all methods (14.2–26.0 per variant), we treat $K$=20 metrics — within every method's budget — as the primary fixed-cutoff comparison and additionally report two set-level quantities: **SetRecall** (recall of the entire returned candidate set) and **Precision@Retrieved** (precision with the actual list length as denominator). An earlier draft of this work reported "Recall@50/Precision@50" computed with the returned-list length as denominator; those quantities are exactly SetRecall and Precision@Retrieved and are named as such here — we do not report any precision at a $K$ exceeding the candidate budget. Aggregates are macro (mean ± population std over 30 variants) and micro (pooled counts over 540 reference papers). Paper identity matching uses normalized DOI, then PMID, then normalized title.

## 4.4 Statistical Protocol

Paired differences are tested with the two-sided Wilcoxon signed-rank test [18] (zero differences dropped; exact null distribution when ranks are tie-free). We declare a single family of eight tests — the component chain and end-to-end comparisons on SetRecall, plus Recall@20, nDCG@50, and Precision@Retrieved endpoints — and control family-wise error with Holm's step-down procedure [19]; all $p$-values below are reported raw and Holm-adjusted. Effect sizes are matched-pairs rank-biserial correlations $r$ (ties excluded), and every paired delta carries a bootstrap 95% confidence interval (10,000 variant-level resamples, fixed seed). Two structural caveats: the M4-vs-M3 comparison is one-sided by construction (a nested extension cannot lower recall), and its test rests on only 3 non-zero pairs; and with $n$=30 variants the component tests are underpowered for deltas of the observed magnitude, so non-significance should not be read as evidence of no effect.

## 4.5 Main Results

*End-to-End Gain Is Significant; Component Attributions Are Preliminary.* Table 4 and Figure 2 summarize the corrected evaluation. End to end, the nested pipeline raises set recall from 0.171 to 0.345 over the PubMed-only baseline — 2.0× the baseline, a paired delta of +0.173 (bootstrap 95% CI [+0.091, +0.251]) that survives Holm correction ($p_{\text{Holm}}$=0.004, $r$=0.79); the fixed-cutoff view agrees (Recall@20 +0.159, $p_{\text{Holm}}$=0.004), as does ranking quality (nDCG@50 0.195 → 0.320, $p_{\text{Holm}}$=0.025, $r$=0.57). Micro-averaged, the pipeline retrieves 193 of 540 reference papers against 83 for the baseline. The component decomposition, however, is weaker than the aggregate: the three step deltas (+0.076 multi-source, +0.082 translation, +0.016 citation graph) sum to +0.173, but after Holm correction none is individually significant at $\alpha$=0.05 (Table 6), and the multi-source CI crosses zero ([−0.017, +0.158]). With $n$=30 variants these tests are underpowered for effects of this size; we therefore describe the decomposition as *observed, near-additive coverage gains* whose per-component significance requires a larger variant set, not as established independent contributions.

**Table 4: Main results (mean ± std over 30 variants; fixed-denominator metrics at $K$=20; set-level metrics over the full returned lists). "Cands" = realized candidates per variant (mean; min–max). "Rel." = pooled relevant found of 540 (micro).**

| Method | Recall@20 | Prec@20 | F1@20 | nDCG@20 | SetRecall | Prec@Retr | Cands | Rel. |
|---|---|---|---|---|---|---|---|---|
| M1 PubMed-only | 0.171 ± 0.183 | 0.138 ± 0.135 | 0.146 ± 0.140 | 0.201 ± 0.172 | 0.171 ± 0.183 | 0.229 ± 0.240 | 14.2 (1–15) | 83 |
| M2 Multi-source | 0.247 ± 0.136 | 0.238 ± 0.160 | 0.239 ± 0.143 | 0.273 ± 0.162 | 0.247 ± 0.136 | 0.238 ± 0.160 | 20.0 (20–20) | 143 |
| M3 +Translation | 0.329 ± 0.099 | 0.303 ± 0.130 | 0.309 ± 0.111 | 0.324 ± 0.117 | 0.329 ± 0.099 | 0.320 ± 0.143 | 19.3 (15–20) | 182 |
| M4 +Cit. graph (nested) | 0.330 ± 0.100 | 0.305 ± 0.133 | 0.311 ± 0.112 | 0.325 ± 0.118 | 0.345 ± 0.114 | 0.261 ± 0.139 | 26.0 (15–40) | 193 |
| M4′ independent re-run | 0.334 ± 0.102 | 0.308 ± 0.134 | 0.314 ± 0.114 | 0.325 ± 0.123 | 0.348 ± 0.116 | 0.264 ± 0.138 | 26.0 | 195 |

For M1–M3 (budgets ≤ 20) Recall@20 coincides with SetRecall by definition. M4′ is the non-nested live re-run, shown only as a run-variance reference (Section 4.6); paper claims use the nested M4.

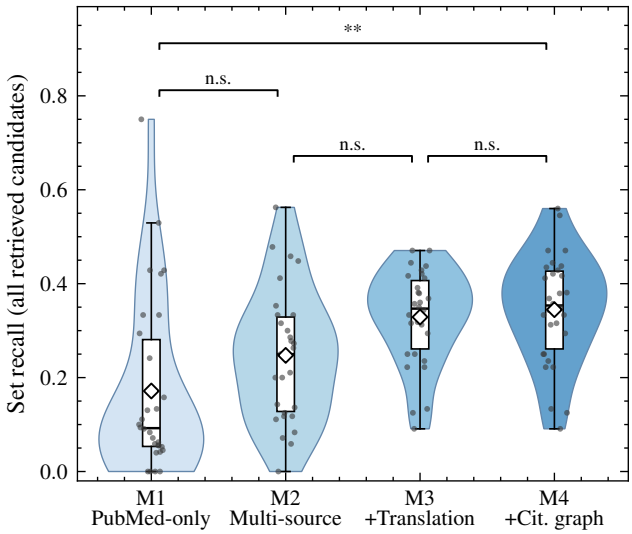



**Figure 2: Ablation study: per-variant set-recall distributions (30 variants). Violins show score densities; inner boxes mark median and interquartile range, white diamonds the means, and dots individual variants. Brackets report paired Wilcoxon tests with *Holm-corrected* $p$-values (∗: $p_{\text{Holm}} < 0.05$; ∗∗: $< 0.01$; n.s.: not significant). M4 is the strictly nested variant.**

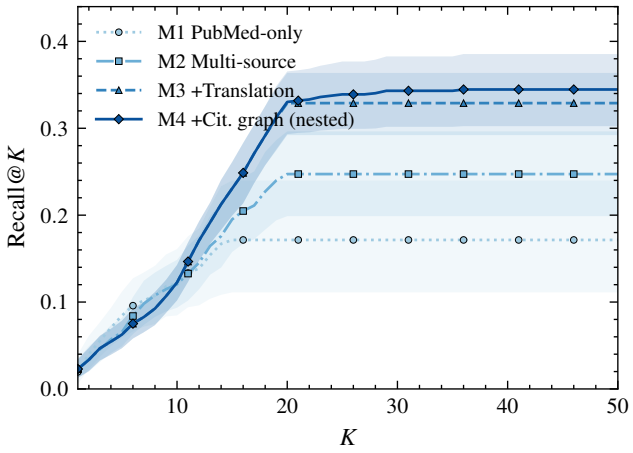



**Figure 3: Mean Recall@$K$ over 30 variants. Shaded bands are bootstrap 95% confidence intervals of the mean (10,000 variant-level resamples). Curves plateau where a method's candidate pool is exhausted (M1: ≤15 papers per variant; M2/M3: 20; nested M4: up to 40). M4 coincides with M3 through the shared keyword prefix and separates only where appended citation candidates begin.**

*Federation Gains Emerge Beyond Shallow Cutoffs.* The Recall@$K$ curves in Figure 3 nearly overlap through $K$=10 (0.120–0.123), with the PubMed baseline marginally ahead at $K$=5 (0.084 vs. 0.063–0.065). PubMed's head ranking is competitive while its candidate pool lasts; separation emerges from candidate-pool depth rather than from the first few ranks. M1's pool self-exhausts at ≤15 papers per variant (plateau 0.171), M2 and M3 saturate at their 20-candidate budget (0.247 and 0.329), and the nested M4 continues to accrue recall until approximately $K$=40 (0.345). The federation advantage on this reference set is thus a property of candidate-set coverage — consistent with the acquisition setting, where the candidate set feeds downstream screening — and the budget asymmetry (14.2 vs. 26 candidates) is itself part of the mechanism: deeper pools mechanically enable higher set recall, which is why fixed-$K$ metrics and explicit budgets accompany every set-level number here.

## 4.6 Run-to-Run API Variance

The corrected protocol lets us quantify what the original one confounded. The M3 run and the keyword phase of the original M4′ run executed the *same* configuration independently, minutes apart; their disagreement directly estimates live-API retrieval variance. Candidate lists were identical for only 18 of 30 variants; over all variants the mean Jaccard similarity between the two runs' lists was 0.891 (minimum 0.429), and the per-variant set recall differed by 0.022 on average (maximum 0.222). Aggregate means were stable (SetRecall 0.329 vs. 0.334 for the keyword phase; 0.345 nested vs. 0.348 for the full M4′), but the per-variant jitter is the same order of magnitude as the component deltas of Sections 4.5 (+0.076/+0.082) and exceeds them on individual variants. Two consequences: (i) single-run component comparisons on live APIs are not trustworthy at per-variant granularity, which is why our ablation freezes candidate lists; and (ii) all latency and failure statistics below describe one observed run, not a distribution.

**Table 5: Per-variant set recall: PubMed-only vs. nested full pipeline (top gainers).**

| Variant | Gene | M1 | M4 |
|---|---|---|---|
| V030 | DMPK | 0.04 | 0.56 |
| V012 | EGFR | 0.05 | 0.55 |
| V011 | KRAS | 0.00 | 0.47 |
| V013 | BRAF | 0.00 | 0.47 |
| V027 | DMD | 0.00 | 0.44 |
| V029 | CYP21A2 | 0.11 | 0.44 |
| V022 | PAH | 0.13 | 0.43 |
| V015 | MLH1 | 0.08 | 0.42 |

**Table 6: Paired Wilcoxon signed-rank tests (30 variants), Holm-corrected as one eight-test family. + = row's second method improves over the first. SR = SetRecall, PR = Precision@Retrieved.**

| Comparison | Δ | 95% CI | $p$ | $p_{\text{Holm}}$ | $r$ |
|---|---|---|---|---|---|
| M2 vs M1, SR | +0.076 | [−0.017, +0.158] | .025 | .102 | +0.50 |
| M3 vs M2, SR | +0.082 | [+0.015, +0.149] | .037 | .102 | +0.45 |
| M4 vs M3, SR[a] | +0.016 | [+0.000, +0.037] | .250 | .250 | +1.00 |
| M4 vs M1, SR | +0.173 | [+0.091, +0.251] | .0005 | .004** | +0.79 |
| M4 vs M2, SR | +0.097 | [+0.026, +0.171] | .028 | .102 | +0.48 |
| M4 vs M1, Recall@20 | +0.159 | [+0.082, +0.230] | .0006 | .004** | +0.78 |
| M4 vs M1, nDCG@50 | +0.124 | [+0.046, +0.199] | .005 | .025* | +0.57 |
| M4 vs M3, PR[b] | −0.058 | [−0.091, −0.029] | .0002 | .002** | −1.00 |

*: $p_{\text{Holm}} < 0.05$; **: $< 0.01$. $r$: matched-pairs rank-biserial correlation (ties excluded).

[a] One-sided by construction (nested extension cannot lower recall); only 3 non-zero pairs.

[b] All 13 non-zero pairs negative by construction (appending citation candidates can only dilute list-level precision).

## 4.7 Per-variant Analysis

*Federation Eliminates Zero-Recall Failures and Compresses Variance.* Beyond shifting the mean, federation stabilizes performance across variants: the cross-variant set-recall std falls from 0.183 (M1) to 0.136 (M2) and 0.099 (M3), rising slightly to 0.114 under the nested M4. The baseline fails outright on four variants (KRAS p.G12D, BRAF p.V600E, APC, and DMD retrieve zero reference papers), whereas the nested pipeline leaves no variant at zero (minimum 0.091). In the paired comparison of Figure 5, the nested pipeline improves 20 of 30 variants, ties 5, and regresses on 5. Per component, multi-source search improves 20 variants (regressing 6), translation 17 (regressing 11), and citation traversal 3 (regressing none, by construction). The largest gains occur where keyword search saturates (DMPK 0.04→0.56, EGFR 0.05→0.55), three of the four baseline-zero variants recover to 0.44–0.47, and APC recovers only to 0.12. The largest regression (MYH7, 0.75→0.25) occurs on a reference set of only 4 papers, where each missed paper changes recall by 0.25; its magnitude therefore also reflects sensitivity to a small denominator.

## 4.8 Statistical Significance

Table 6 reports the declared eight-test family with raw and Holm-adjusted $p$-values, effect sizes, and bootstrap CIs.

*Citation Traversal Trades List-Level Precision for Otherwise Unreachable Recall.* Under fixed-denominator precision the ablation chain is monotone non-decreasing (Prec@20: 0.138 → 0.238 → 0.303 → 0.305), so the often-cited "precision drop" of citation augmentation exists only in the list-level view: Precision@Retrieved falls from 0.320 (M3) to 0.261 (M4) because the appended citation candidates are mostly non-relevant (11 of 202 appended items, 5.4%, match the reference set, against 43.2% for Crossref-attributed and 22.5% for PMC-attributed keyword candidates). The trade is asymmetric in kind: 10 of the 11 relevant citation-graph papers appear in *no* keyword-channel list recorded across any of our runs, and they concentrate on three variants (EGFR 5, DMPK 5, PAH 1); F1@20 is essentially unchanged (0.309 → 0.311). A screening-oriented configuration may prefer M3; recall-oriented evidence acquisition, where a missed paper is costlier than an extra screened candidate, favors enabling traversal — with the caveat that the recall benefit reached only 3 of 30 variants in this reference set.

## 4.9 Source Attribution, Channel Overlap, and Fusion

*First-Seen Attribution Is Not Source Complementarity.* Figure 6 reports per-source candidate counts for the nested pipeline: Crossref-labeled records contribute 131 of the 193 relevant papers (67.9%), PMC 49, citation graph 11, EuropePMC 2. These labels are *first-seen attribution*: deduplication keeps the first provider that surfaced each record and discards later occurrences, so per-source labels are disjoint *by construction* and cannot support claims that the sources' underlying indexes do not overlap (an earlier draft presented a zero-overlap source matrix as evidence of complementarity; that reading was invalid and is withdrawn). Measuring true index overlap requires per-source raw streams, which the current pipeline does not persist.

*Method-Level Overlap Reveals Channel Disjointness.* Overlap *between methods* (separate runs, no shared deduplication) is measurable and informative. The nested pipeline and the PubMed baseline share only 12 relevant (variant, paper) pairs: M4 finds 181 relevant papers M1 misses, but M1 also finds 71 relevant papers M4 misses — 39.7% of everything federation retrieved is invisible to PubMed-only search and vice versa for 85.5% of PubMed's yield. The mechanism is configuration, not magic: the `auto` routing plan queries PMC, Crossref, OpenAlex, EuropePMC, etc., but not the PubMed E-utilities channel itself (Table 3), and with 20-candidate budgets the federated ranking surfaces different papers than PubMed's recency-ranked list. Across all four methods the union reaches 354 of 540 reference papers (65.6%) — far above any single method — indicating that under tight budgets every evaluated configuration leaves substantial reference coverage on the table.

*Transparent Fusion Baselines.* The reviewer-facing question "is federation better than simply combining channels?" can be answered offline from the recorded runs (Table 7). A simple union of the M1 and M4 candidate sets lifts set recall to 0.493 macro (264/540 micro) — +0.148 over the nested pipeline alone — at the cost of a larger screening set (≈40 candidates per variant). Reciprocal Rank Fusion [20] ($k$=60) over the two ranked lists reaches the same set-level coverage but *degrades* early precision relative to M4 alone

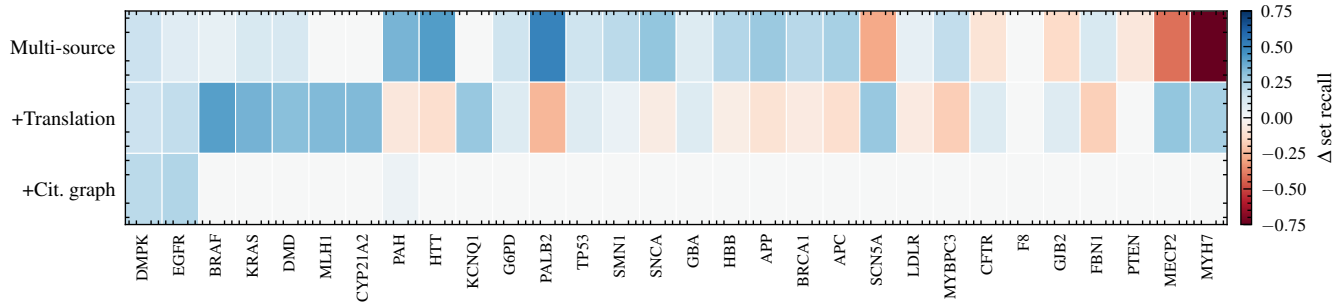



**Figure 4: Per-variant set-recall change contributed by each pipeline component under the nested protocol (columns sorted by total M1→M4 gain). Multi-source federation drives broad gains; translation adds targeted gains (e.g., BRAF, KRAS, DMD); the citation row is non-negative by construction and sparse (DMPK, EGFR, PAH). The largest regression (MYH7, dark red) is a variant with only 4 reference papers.**

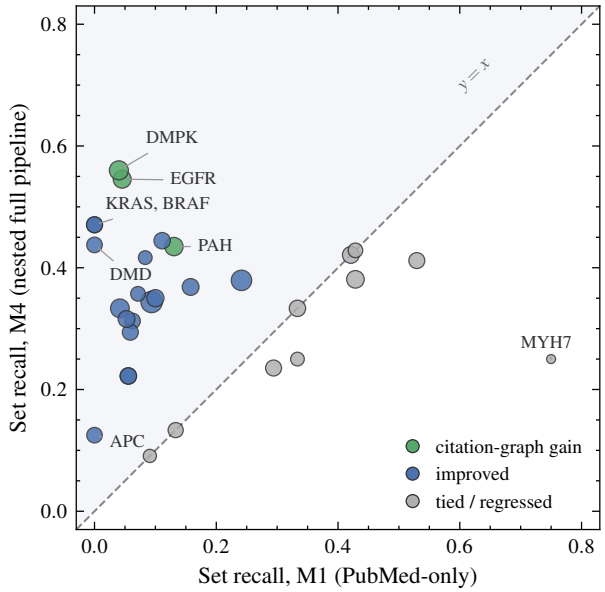



**Figure 5: Per-variant set recall: nested full pipeline (M4) vs. PubMed-only (M1). Points above the $y{=}x$ line improve under the full pipeline (20/30; 5 tied, 5 regress). Marker area is proportional to the variant's reference-set size (4–32 papers); green marks the variants with citation-graph contributions (EGFR, DMPK, PAH). KRAS and BRAF coincide at $(0, 0.47)$.**

(Recall@20 0.231 vs. 0.330; Prec@20 0.212 vs. 0.305): because the two channels are nearly disjoint, interleaving them dilutes the head of the ranking with the weaker channel's non-relevant items rather than promoting corroborated ones. Practical implication: include the PubMed channel in the acquisition plan and fuse at the *set* level for screening workflows, rather than rank-interleaving disjoint channels.

**Table 7: Offline fusion of the recorded M1 and nested M4 runs.**

| Configuration | Recall@20 | Prec@20 | SetRecall | Rel. (540) |
|---|---|---|---|---|
| M1 alone | 0.171 | 0.138 | 0.171 | 83 |
| M4 alone (nested) | 0.330 | 0.305 | 0.345 | 193 |
| RRF(M1, M4), $k$=60 | 0.231 | 0.212 | 0.493 | 264 |
| Union(M1, M4) | — | — | 0.493 | 264 |

Union is order-free (no ranking), hence no fixed-$K$ metrics.

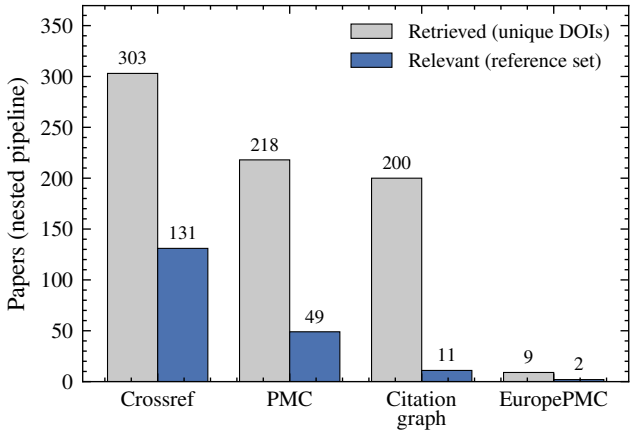



**Figure 6: Per-source candidates for the nested pipeline under *first-seen attribution* (deduplication keeps the first provider that returned each record): unique DOIs retrieved vs. reference-set relevant. Labels do not measure exclusive availability.**

## 4.10 Multilingual Contribution (German and French Only)

The reference set contains 24 non-English papers (13 German, 11 French); no other language carries annotations, so we make no

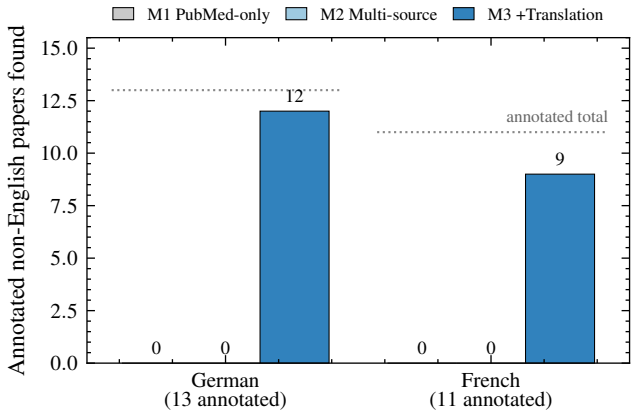



**Figure 7: Annotated German/French reference papers recovered per method (annotation-side language labels; dotted caps mark annotated totals). M2 vs. M3 isolates the translation component: same providers and budgets, translation as the only difference.**

claims about the remaining target languages (zh, ja, ru). The controlled contrast for the translation component is M2 vs. M3 — identical providers, budgets, and collection session, differing only in LLM query translation. As Figure 7 shows, using the *annotation-side* language labels (robust to missing retrieval metadata): M1 and M2 recover 0 of 24; M3 recovers 21 of 24 (12/13 German, 9/11 French). English-only retrieval misses the annotated non-English evidence entirely — a structural gap, not a ranking difference — and translation with language-aware routing closes most of it. Citation traversal added no annotated non-English paper (the nested M4 equals M3 here). We note for transparency that the non-nested M4′ rerun happened to recover the two remaining French papers (23/24); under the frozen-list protocol that gain is attributable to keyword re-retrieval variance, not to any pipeline component — an example of how non-nested comparisons overstate component effects. Because 228 reference papers lack language metadata, per-language totals are conservative lower bounds on non-English content.

## 4.11 Candidate List Integrity

What the recorded lists let us verify about deduplication is narrower than a precision/recall evaluation, and we state it as such. Within-variant residual duplicates (same DOI/PMID/normalized title appearing twice in one returned list) are **zero** across all 2,384 recorded entries of all methods, i.e., the multi-key procedure left no visible duplicates in its output streams. Records lacking a DOI are rare (M1: 5 of 426, 1.2%; M2: 4 of 600; M3: 0 of 578; M4: 1 of 780). The same paper legitimately recurs across *different* variants (8–49 identities per method), which is expected topical overlap, not duplication failure. An earlier draft reported "within-method DOI duplication of 2.8–8.8%"; those figures conflated missing-DOI records and legitimate cross-variant repetition with duplication and are withdrawn. Deduplication precision/recall against ground-truth duplicate pairs requires the raw pre-deduplication provider streams, which the collection runs did not persist; instrumenting the pipeline to retain them is future work. A 50-pair manual audit found no false merges, but we report this as anecdotal (not a systematic sample) and draw no quantitative conclusion from it.

**Table 8: External single-source baselines (English query, budget 50, default API ranking, collected 2026-08-13) vs. recorded pipeline runs.**

| Configuration | Cands | Recall@20 | Prec@20 | SetRecall | Rel. (540) |
|---|---|---|---|---|---|
| EuropePMC direct | 48.5 | 0.110 | 0.093 | 0.140 ± 0.114 | 72 |
| OpenAlex direct | 48.5 | 0.047 | 0.040 | 0.077 ± 0.068 | 41 |
| Crossref direct | 50.0 | 0.141 | 0.118 | 0.210 ± 0.146 | 109 |
| RRF(3 externals) | — | 0.199 | 0.170 | 0.381 | 197 |
| Union(3 externals) | — | — | — | 0.381 | 197 |
| M1 PubMed-only (2026-08-12) | 14.2 | 0.171 | 0.138 | 0.171 ± 0.183 | 83 |
| M4 nested (2026-08-12) | 26.0 | 0.330 | 0.305 | 0.345 ± 0.114 | 193 |

Semantic Scholar direct: unavailable under anonymous rate limits (29/30 queries failed); recorded, not tabulated. External runs postdate the main runs by one day.

## 4.12 External Single-Source Baselines and Reference-Set Bias

To situate the pipeline against retrieval channels outside our own stack, we queried three public APIs directly — EuropePMC, OpenAlex, and Crossref REST — with each variant's English query, a 50-candidate budget, and each API's default relevance ranking (no LLM, no routing, no fusion); anonymous Semantic Scholar access was rate-limited into unavailability (29 of 30 queries failed after retries) and is reported as such. These runs were collected on 2026-08-13, one day after the main runs, so cross-snapshot comparisons carry that caveat. Table 8 shows that no naive single channel approaches the pipeline: the best (Crossref direct) reaches set recall 0.210 with 50 candidates per variant, versus 0.345 for the nested pipeline with 26. Fusing all three external channels (union or RRF) reaches 0.381 — above any single method but still below the recorded M1∪M4 union (0.493) at comparable total budget. Two readings follow. First, reference-set coverage is not trivially saturated by re-querying the APIs that seeded the reference pool: query formulation, ranking, and routing matter, so the pooling bias of Section 4.1, while real, does not reduce the benchmark to an identity check. Second, none of the external English-query channels recovered a single annotated German or French paper, corroborating that the non-English gap is structural and closed specifically by translation-aware retrieval (Section 4.10).

## 4.13 Acquisition Latency

*Observed Wall-Clock Cost Is Rate-Limit Dominated.* Table 9 reports observed single-run wall-clock times: mean acquisition rises from 3.1 s per variant for PubMed-only to 50.7 ± 11.8 s for the original full-pipeline run (median 46.5 s, range 37.1–82.5 s). The non-monotonic ordering of the federated configurations — M2 at 81.7 s per variant exceeds M3 (63.3 s) and M4′ (50.7 s) despite executing fewer stages — together with the 78 rate-limit rejections in the log, shows that these times measure end-to-end latency under external API throttling, not the marginal computational cost of translation or traversal, and single-run values carry the run-to-run variance of Section 4.6. The nested M4 is a derived configuration (frozen M3 list + traversal); its incremental cost is the citation-traversal phase,

**Table 9: Acquisition wall-clock time (30 variants, single run, sequential execution).**

| Method | Papers | Total (s) | Per variant (s) |
|---|---|---|---|
| M1 PubMed-only | 426 | 92 | 3.1 |
| M2 Multi-source | 600 | 2452 | 81.7 |
| M3 +Translation | 578 | 1898 | 63.3 |
| M4′ full re-run | 780 | 1522 | 50.7 ± 11.8 |

M4′ dispersion from per-variant measurements (median 46.5 s, range 37.1–82.5 s); M1–M3 report batch totals divided by 30. Times include provider rate-limit backoff, which dominates the federated methods and varies across runs; the corrected benchmark runner records per-variant timings for every method.

bounded above by the recorded M4′ per-variant times. All methods retain sub-minute median acquisition per variant.

## 4.14 Case Study: Citation Graph Recall

The citation graph contributes where keyword search is saturated, but on this reference set the effect is narrow: 3 of 30 variants (Figure 8). Two variants illustrate the mechanism:

**EGFR p.L858R (V012, lung adenocarcinoma).** Five reference papers are found *exclusively* via citation traversal, including a 2024 JAMA randomized trial (Ivonescimab plus chemotherapy in EGFR-mutated NSCLC, DOI 10.1001/jama.2024.10613) and a 2023 Lancet Respiratory Medicine trial (Sintilimab plus chemotherapy). These are recent, high-impact clinical papers whose titles do not surface under the variant-keyword query used by M1–M3. We note they also exemplify the reference-set validity concern of Section 4.1: treatment trials support clinical management rather than variant classification per se, so their annotation as "relevant" reflects the keyword protocol, not ACMG evidence review.

**DMPK CTG repeat (V030, myotonic dystrophy type 1).** Citation traversal recovers 5 reference papers missed by keyword search, including a 2019 *Neurobiology of Disease* review of congenital myotonic dystrophy molecular genetics (10.1016/j.nbd.2019.104533) and a 2022 *Genes* review (10.3390/genes13020367) — canonical works connected by citation edges but not by variant-name lexical match.

## 4.15 Reproducibility and Released Artifacts

We release, alongside the MIT-licensed system code: the reference set (30 variants, 540 annotations with language and evidence-type fields); the raw per-method ranked candidate lists exactly as collected (2026-08-12, 12:09–13:47 local, single session); the collection log (78 Semantic Scholar HTTP-429 events, 120 J-STAGE parse failures, 60 translation calls); the translation system prompt (temperature 0.2, six target languages) whose SHA-256 fingerprint is embedded in the released run manifest; and a deterministic analysis script (`make_paper_data.py`) that regenerates every table, figure, and statistic in this paper from the raw lists, including the nested-M4 construction, the Holm-corrected test family, and bootstrap CIs (10,000 resamples, fixed seed 42). The benchmark runner records a machine-readable manifest (configuration echo, model identifiers, per-variant timings, budgets) for future collection runs; the

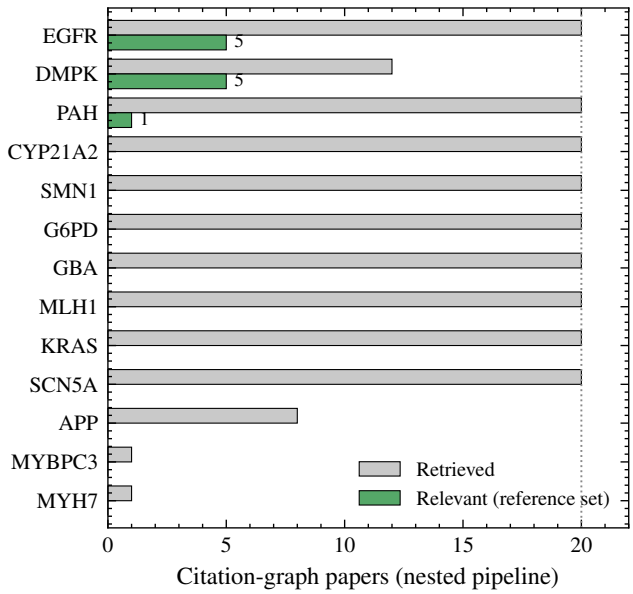



**Figure 8: Citation graph traversal per gene (nested pipeline): papers retrieved vs. reference-set relevant. The dotted line marks the traversal budget (20 papers per variant); relevant contributions concentrate in EGFR (5), DMPK (5), and PAH (1).**

LLM endpoint used for the 2026-08-12 translation run was configured via environment and its identifier was not captured by the original log — a reproducibility gap we document rather than obscure, and which the manifest now closes for subsequent runs. All retrieval-dependent results reflect live-API state in the stated collection window.

# 5 DISCUSSION

**Limitations.** In order of severity: (i) *Reference-set construct validity is the binding constraint on every conclusion*: the annotations are heuristic, expert-unverified, and drawn from a candidate pool whose channels overlap the evaluated system (Section 4.1); metrics are relative coverage of that pool, and genuinely clinical claims (e.g., PS3/PVS1 evidence retrieval quality) are out of reach of this evaluation. (ii) Candidate budgets in the recorded runs were 14.2–26 per variant (client caps and rate limits), below the $K$=50 originally advertised; we therefore report fixed-$K$ metrics only within budget and label set-level metrics as such, and the corrected runner defaults to uniform 50-candidate budgets. (iii) All online results are single runs; the measured run-to-run jitter (Section 4.6) is of the same order as the component deltas, so per-component conclusions are preliminary. (iv) The relevance gate, OA download layer, health-aware scheduling, and deduplication accuracy are engineering features without dedicated experiments here (gating accuracy/cost, download success and license correctness, fault-injection behavior, and dedup precision/recall are each unmeasured). (v) Language metadata is missing for 228 reference papers, and only German/French

carry non-English annotations. (vi) The external baselines of Section 4.12 were collected one day after the main runs; cross-snapshot comparisons carry that caveat.

**Future work.** Expert re-annotation is the priority: dual independent annotation with clinical-genetics expertise on a stratified sample of pooled candidates (including candidates from systems that do not share our providers), adjudication, and reported inter-annotator agreement (Cohen's $\kappa$), distinguishing "about the gene/disease" from "usable as variant-level evidence". Protocol-side: a full re-run under the corrected uniform-budget nested protocol; per-language translation fidelity evaluation (gene-symbol and HGVS retention rates, now measurable since the runner persists translated queries); dedicated experiments for the gate (precision/recall/cost on human labels), download layer (OA resolution success, license correctness, valid-PDF rate), health scheduler (fault injection), and deduplication (precision/recall on retained raw streams); and embedding-based semantic retrieval as an additional recall channel.

## 6 CONCLUSION

We presented lit-acquisition, an open-source multi-source multilingual biomedical literature acquisition system, together with an exploratory, heuristically annotated reference set and a corrected evaluation protocol (strictly nested ablation, budget-honest fixed-$K$ metrics, Holm-corrected paired statistics, quantified run-to-run API variance, and transparent fusion baselines). On this reference set the end-to-end pipeline roughly doubles PubMed-only set recall (0.171 $\rightarrow$ 0.345, Holm-adjusted $p$=0.004); individual component contributions are positive but not individually significant at $n$=30, translation recovers 21 of 24 annotated German/French papers in a controlled contrast, and citation traversal adds a small set of otherwise-unretrieved papers on 3 variants at a significant list-level precision cost. We position this work as infrastructure — code, unified provider interfaces, raw run data, and a deterministic analysis pipeline — for the community to build clinically validated evaluations on, not as evidence of clinical retrieval quality; the reference set requires expert re-annotation before any such claim.

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